

Nicolás Garavito-Camargo – Publications

Publications list

Refereed: 36 – First author: 5 – Supervised students: 3 (denoted by †) – Total citations: 1655 (as of June 01, 2026)

[ORCID](#), [ADS](#), [arXiv](#), [Google Scholar](#)

In press

3. † *Shaping the Milky Way. II. The dark matter halo’s response to the LMC’s passage in a cosmological context*
Darragh-Ford, Elise; **Garavito-Camargo, Nicolas**; Arora, Arpit; Wechsler, Risa H.; Mansfield, Phil; Besla, Gurtina; Petersen, Michael S.; Weinberg, Martin D.; Varela-Lavin, Silvio; Buch, Deveshi; et al..
arXiv e-prints, November 2025. [5 citations]
2. *The Milky Way - Large Magellanic Cloud Interaction with Simulation Based Inference*
Brooks, Richard A. N.; Sanders, Jason L.; Chandra, Vedant; **Garavito-Camargo, Nicolás**; Dillamore, Adam M.; Price-Whelan, Adrian M.; Ting, Yuan-Sen.
arXiv e-prints, October 2025. [7 citations]
1. *Where do High-Velocity Dark Matter Particles come from in the Milky Way?*
DeBrae, Aidan; Behroozi, Peter; **Garavito-Camargo, Nicolas**.
arXiv e-prints, February 2025. [3 citations]

First author and student-led publications

8. † *Shaping the Milky Way: The Interplay of Mergers and Cosmic Filaments*
Arora, Arpit; **Garavito-Camargo, Nicolás**; Sanderson, Robyn E.; Weinberg, Martin D.; Petersen, Michael S.; Varela-Lavin, Silvio; Gómez, Facundo A.; Johnston, Kathryn V.; Laporte, Chervin F. P.; Shipp, Nora; et al..
The Astrophysical Journal, August 2025. [20 citations]
7. † *LMC Calls, Milky Way Halo Answers: Disentangling the Effects of the MW-LMC Interaction on Stellar Stream Populations*
Brooks, Richard A. N.; **Garavito-Camargo, Nicolás**; Johnston, Kathryn V.; Price-Whelan, Adrian M.; Sanders, Jason L.; Lilleengen, Sophia.
The Astrophysical Journal, January 2025. [19 citations]
6. *On the Corotation of Milky Way Satellites: LMC-mass Satellites Induce Apparent Motions in Outer Halo Tracers*
Garavito-Camargo, Nicolás; Price-Whelan, Adrian M.; Samuel, Jenna; Cunningham, Emily C.; Patel, Ekta; Wetzel, Andrew; Johnston, Kathryn V.; Arora, Arpit; Sanderson, Robyn E.; Garrison, Lehman; et al..
The Astrophysical Journal, November 2024. [19 citations]
5. † *LMC-driven Anisotropic Boosts in Stream-Subhalo Interactions*
Arora, Arpit; **Garavito-Camargo, Nicolás**; Sanderson, Robyn E.; Cunningham, Emily C.; Wetzel, Andrew; Panithanpaisal, Nondh; Barry, Megan.
The Astrophysical Journal, October 2024. [20 citations]
4. *The Clustering of Orbital Poles Induced by the LMC: Hints for the Origin of Planes of Satellites*
Garavito-Camargo, Nicolás; Patel, Ekta; Besla, Gurtina; Price-Whelan, Adrian M.; Gómez, Facundo A.; Laporte, Chervin F. P.; Johnston, Kathryn V..
The Astrophysical Journal, December 2021. [39 citations]
3. *Quantifying the Impact of the Large Magellanic Cloud on the Structure of the Milky Way’s Dark Matter Halo Using Basis Function Expansions*
Garavito-Camargo, Nicolás; Besla, Gurtina; Laporte, Chervin F. P.; Price-Whelan, Adrian M.; Cunningham, Emily C.; Johnston, Kathryn V.; Weinberg, Martin; Gómez, Facundo A..
The Astrophysical Journal, October 2021. [133 citations]

2. *Hunting for the Dark Matter Wake Induced by the Large Magellanic Cloud*
Garavito-Camargo, Nicolas; Besla, Gurtina; Laporte, Chervin F. P.; Johnston, Kathryn V.; Gómez, Facundo A.; Watkins, Laura L.
The Astrophysical Journal, October 2019. [221 citations]
1. *The Impact of Gas Bulk Rotation on the Ly α Line*
Garavito-Camargo, Juan N.; Forero-Romero, Jaime E.; Dijkstra, Mark.
The Astrophysical Journal, November 2014. [11 citations]

Publications as collaborator

28. *Quantifying the Milky Way, LMC and their interaction using all-sky kinematics of outer halo stars*
Brooks, Richard A. N.; Sanders, Jason L.; Dillamore, Adam M.; Garavito-Camargo, Nicolás; Chandra, Vedant; Price-Whelan, Adrian M.; Cargile, Phillip.
Monthly Notices of the Royal Astronomical Society, May 2026. [1 citations]
27. *Stellar Birth Radii in the LMC: Insights into Chemodynamics, Radial Migration, and Star Formation across the Disk*
Lu, Yuxi(Lucy); Garver, Bethany; Nidever, David L.; Povick, Joshua T.; Garavito-Camargo, Nicolás; Cioni, Maria-Rosa L.; Horta, Danny.
The Astronomical Journal, April 2026. [1 citations]
26. *Mapping the Distorted Dark Matter Distribution of the LMC–SMC System Prior to Milky Way Infall with Basis Function Expansions*
Foote, Hayden R.; Rathore, Himansh; Besla, Gurtina; Garavito-Camargo, Nicolás; Patel, Ekta; Petersen, Michael S.; Weinberg, Martin D.; Gómez, Facundo A.; Laporte, Chervin F. P.; EXP Collaboration.
The Astrophysical Journal, April 2026. [2 citations]
25. *A simulation-based inference approach to the dynamics of the MW–LMC system - validation*
Brooks, Richard A. N.; Sanders, Jason L.; Dillamore, Adam M.; Garavito-Camargo, Nicolás; Price-Whelan, Adrian M..
Monthly Notices of the Royal Astronomical Society, January 2026. [12 citations]
24. *All-sky Kinematics of the Distant Halo: The Reflex Response to the LMC*
Chandra, Vedant; Naidu, Rohan P.; Conroy, Charlie; Garavito-Camargo, Nicolas; Laporte, Chervin; Bonaca, Ana; Cargile, Phillip A.; Cunningham, Emily; Han, Jiwon Jesse; Johnson, Benjamin D.; et al..
The Astrophysical Journal, August 2025. [36 citations]
23. *The M31–M33 Interaction: Impact on M31’s Center-of-mass Motion and Satellite Orbits*
Patel, Ekta; Garavito-Camargo, Nicolás; Escala, Ivanna.
The Astrophysical Journal, May 2025. [5 citations]
22. *Hypervelocity Stars Trace a Supermassive Black Hole in the Large Magellanic Cloud*
Han, Jiwon Jesse; El-Badry, Kareem; Lucchini, Scott; Hernquist, Lars; Brown, Warren; Garavito-Camargo, Nico; Conroy, Charlie; Sari, Re'em.
The Astrophysical Journal, April 2025. [22 citations]
21. *The Distant Milky Way Halo from the Southern Hemisphere: Characterization of the LMC-induced Dynamical Friction Wake*
Cavieres, Manuel; Chanamé, Julio; Navarrete, Camila; Ordenes-Briceño, Yasna; Garavito-Camargo, Nicolás; Besla, Gurtina; Hempel, Maren; Vivas, A. Katherina; Gómez, Facundo.
The Astrophysical Journal, April 2025. [12 citations]
20. *The All-sky Impact of the LMC on the Milky Way Circumgalactic Medium*
Carr, Christopher; Bryan, Greg L.; Garavito-Camargo, Nicolás; Besla, Gurtina; Setton, David J.; Johnston, Kathryn V.; Su, Kung-Yi.
The Astrophysical Journal, April 2025. [7 citations]
19. *Segue 2 Recently Collided with the Cetus-Palca Stream: New Opportunities to Constrain Dark Matter in an Ultra-faint Dwarf*
Foote, Hayden R.; Besla, Gurtina; Garavito-Camargo, Nicolás; Patel, Ekta; Thomas, Guillaume F.; Bonaca, Ana; Price-Whelan, Adrian M.; Peter, Annika H. G.; Zaritsky, Dennis; Conroy, Charlie.
The Astrophysical Journal, February 2025. [3 citations]

18. *Efficient and Accurate Force Replay in Cosmological-baryonic Simulations*
Arora, Arpit; Sanderson, Robyn; Regan, Christopher; **Garavito-Camargo, Nicolás**; Bregou, Emily; Panithanpaisal, Nondh; Wetzel, Andrew; Cunningham, Emily C.; Loebman, Sarah R.; Dropulic, Adriana; et al..
The Astrophysical Journal, December 2024. [18 citations]
17. *Generating synthetic star catalogs from simulated data for next-gen observatories with py-ananke*
Thob, Adrien; Sanderson, Robyn; Eden, Andrew; Nikakhtar, Farnik; Panithanpaisal, Nondh; **Garavito-Camargo, Nicolás**; Sharma, Sanjib.
The Journal of Open Source Software, October 2024. [7 citations]
16. *Dark Matter Distribution in Milky Way analog Galaxies*
de Isidio, Natanael G.; Menéndez-Delmestre, K.; Gonçalves, T. S.; Grossi, M.; Rodrigues, D. C.; **Garavito-Camargo, N.**; Araujo-Carvalho, A.; Beaklini, P. P. B.; Cavalcante-Coelho, Y.; Cortesi, A.; et al..
The Astrophysical Journal, August 2024. [10 citations]
15. *Galactoseismology in cosmological simulations. Vertical perturbations by dark matter, satellite galaxies, and gas*
García-Conde, B.; Antoja, T.; Roca-Fàbrega, S.; Gómez, F.; Ramos, P.; **Garavito-Camargo, N.**; Gómez-Flechoso, M. A..
Astronomy and Astrophysics, March 2024. [14 citations]
14. *The proto-galaxy of Milky Way-mass haloes in the FIRE simulations*
Horta, Danny; Cunningham, Emily C.; Sanderson, Robyn; Johnston, Kathryn V.; Deason, Alis; Wetzel, Andrew; McCluskey, Fiona; **Garavito-Camargo, Nicolás**; Necib, Lina; Faucher-Giguère, Claude-André; et al..
Monthly Notices of the Royal Astronomical Society, February 2024. [28 citations]
13. *The impact of the Large Magellanic Cloud on dark matter direct detection signals*
Smith-Orlik, Adam; Ronaghi, Nima; Bozorgnia, Nassim; Cautun, Marius; Fattahi, Azadeh; Besla, Gurtina; Frenk, Carlos S.; **Garavito-Camargo, Nicolás**; Gómez, Facundo A.; Grand, Robert J. J.; et al..
Journal of Cosmology and Astroparticle Physics, October 2023. [32 citations]
12. *Structure, Kinematics, and Observability of the Large Magellanic Cloud's Dynamical Friction Wake in Cold versus Fuzzy Dark Matter*
Foote, Hayden R.; Besla, Gurtina; Mocz, Philip; **Garavito-Camargo, Nicolás**; Lancaster, Lachlan; Sparre, Martin; Cunningham, Emily C.; Vogelsberger, Mark; Gómez, Facundo A.; Laporte, Chervin F. P..
The Astrophysical Journal, September 2023. [28 citations]
11. *Lopsided galaxies in a cosmological context: a new galaxy-halo connection*
Varela-Lavin, Silvio; Gómez, Facundo A.; Tissera, Patricia B.; Besla, Gurtina; **Garavito-Camargo, Nicolás**; Marinacci, Federico; Laporte, Chervin F. P..
Monthly Notices of the Royal Astronomical Society, August 2023. [17 citations]
10. *Implications of the Milky Way Travel Velocity for Dynamical Mass Estimates of the Local Group*
Chamberlain, Katie; Price-Whelan, Adrian M.; Besla, Gurtina; Cunningham, Emily C.; **Garavito-Camargo, Nicolás**; Peñarrubia, Jorge; Petersen, Michael S..
The Astrophysical Journal, January 2023. [32 citations]
9. *On the Stability of Tidal Streams in Action Space*
Arora, Arpit; Sanderson, Robyn E.; Panithanpaisal, Nondh; Cunningham, Emily C.; Wetzel, Andrew; **Garavito-Camargo, Nicolás**.
The Astrophysical Journal, November 2022. [41 citations]
8. *All-sky dynamical response of the Galactic halo to the Large Magellanic Cloud*
Conroy, Charlie; Naidu, Rohan P.; **Garavito-Camargo, Nicolás**; Besla, Gurtina; Zaritsky, Dennis; Bonaca, Ana; Johnson, Benjamin D..
Nature, April 2021. [138 citations]
7. *Quantifying the Stellar Halo's Response to the LMC's Infall with Spherical Harmonics*
Cunningham, Emily C.; **Garavito-Camargo, Nicolas**; Deason, Alis J.; Johnston, Kathryn V.; Erkal, Denis; Laporte, Chervin F. P.; Besla, Gurtina; Luger, Rodrigo; Sanderson, Robyn E..
The Astrophysical Journal, July 2020. [66 citations]
6. *The Orbital Histories of Magellanic Satellites Using Gaia DR2 Proper Motions*
Patel, Ekta; Kallivayalil, Nitya; **Garavito-Camargo, Nicolas**; Besla, Gurtina; Weisz, Daniel R.; van der Marel, Roeland P.; Boylan-Kolchin, Michael; Pawlowski, Marcel S.; Gómez, Facundo A..
The Astrophysical Journal, April 2020. [197 citations]

5. *The highest-speed local dark matter particles come from the Large Magellanic Cloud*
Besla, G.; Peter, A. H. G.; **Garavito-Camargo, N.**
Journal of Cosmology and Astroparticle Physics, November 2019. [46 citations]
4. *The influence of Sagittarius and the Large Magellanic Cloud on the stellar disc of the Milky Way Galaxy*
Laporte, Chervin F. P.; Johnston, Kathryn V.; Gómez, Facundo A.; **Garavito-Camargo, Nicolas**; Besla, Gurtina.
Monthly Notices of the Royal Astronomical Society, November 2018. [232 citations]
3. *The Extremely Luminous Quasar Survey in the Sloan Digital Sky Survey Footprint. II. The North Galactic Cap Sample*
Schindler, Jan-Torge; Fan, Xiaohui; McGreer, Ian D.; Yang, Jinyi; Wang, Feige; Green, Richard; **Garavito-Camargo, Nicolas**; Huang, Yun-Hsin; O'Donnell, Christine; Patej, Anna; et al..
The Astrophysical Journal, August 2018. [21 citations]
2. *Modelling the gas kinematics of an atypical Ly α emitting compact dwarf galaxy*
Forero-Romero, Jaime E.; Gronke, Max; Remolina-Gutiérrez, Maria Camila; **Garavito-Camargo, Nicolás**; Dijkstra, Mark.
Monthly Notices of the Royal Astronomical Society, February 2018. [2 citations]
1. *Response of the Milky Way's disc to the Large Magellanic Cloud in a first infall scenario*
Laporte, Chervin F. P.; Gómez, Facundo A.; Besla, Gurtina; Johnston, Kathryn V.; **Garavito-Camargo, Nicolas**.
Monthly Notices of the Royal Astronomical Society, January 2018. [143 citations]

White papers

4. *NANCY: Next-generation All-sky Near-infrared Community survey*
Han, Jiwon Jesse; Dey, Arjun; Price-Whelan, Adrian M.; Najita, Joan; Schlafly, Edward F.; Saydjari, Andrew; Wechsler, Risa H.; Bonaca, Ana; Schlegel, David J; Conroy, Charlie; et al..
arXiv e-prints, June 2023. [9 citations]
3. *From Data to Software to Science with the Rubin Observatory LSST*
Breivik, Katelyn; Connolly, Andrew J.; Ford, K. E. Saavik; Jurić, Mario; Mandelbaum, Rachel; Miller, Adam A.; Norman, Dara; Olsen, Knut; O'Mullane, William; Price-Whelan, Adrian; et al..
arXiv e-prints, August 2022. [15 citations]
2. *Mass Spectroscopy of the Milky Way*
Dey, Arjun; Najita, Joan R.; Koposov, Sergey; Rockosi, Connie; Li, Ting; Olsen, Knut; Allende Prieto, Carlos; Bonaca, Ana; Besla, Gurtina; **Garavito-Camargo, Nicolas**; et al..
Bulletin of the American Astronomical Society, May 2019. [1 citations]
1. *The Multidimensional Milky Way*
Sanderson, Robyn; Carlin, Jeff; Cunningham, Emily; **Garavito-Camargo, Nicolas**; Guhathakurta, Puragra; Johnston, Kathryn; Laporte, Chervin; Li, Ting; Sohn, S. Tony; Anderson, Jay; et al..
Bulletin of the American Astronomical Society, May 2019. [8 citations]